

# Task 04 — Account Tier & Rating Automation

## *Detailed Project Report — Step-by-Step Code Walkthrough*

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**Scope:** The 5-file minimal solution — AccountTriggerHandler.cls, AccountTrigger.trigger, AccountTierBatch.cls, AccountTriggerHandlerTest.cls, AccountTierBatchTest.cls

**Working dataset:** 100,000 Account records imported from salesforce\_accounts\_100k.csv into a scratch org

This report walks through every method in the solution block-by-block — what it does, why it's written that way, and which Apex concept it demonstrates — rather than just listing the code. It closes with how to run everything from Developer Console against your imported 100,000-record dataset, and the verification queries to run afterwards.

# 1. Architecture at a Glance

Three moving pieces, five files:

| File                          | Runs when...                                  | Job                                                      |
|-------------------------------|-----------------------------------------------|----------------------------------------------------------|
| AccountTriggerHandler.cls     | Called by the trigger, on every insert/update | Holds the Tier/Rating business rule AND decides when     |
| AccountTrigger.trigger        | Automatically, on Account insert/update       | Routes to the handler. Contains no logic of its own.     |
| AccountTierBatch.cls          | Manually invoked once (or on a schedule)      | Sweeps every EXISTING Account and backfills Tier/Rat     |
| AccountTriggerHandlerTest.cls | Setup → Apex Test Execution                   | Proves the real-time logic (single record + 200-record b |
| AccountTierBatchTest.cls      | Setup → Apex Test Execution                   | Proves the batch correctly fixes a mixed "already correc |

Data flow for your scratch org specifically: the CSV import inserted 100,000 raw Accounts. If AccountTrigger was active during that import, those records already have correct Tier/Rating (the trigger fired on every insert batch the Data Loader sent). If it wasn't active yet, they're blank until AccountTierBatch runs. Either way, the batch is idempotent — running it again does nothing to records that are already correct (see §3's change-detection logic).

## 2. AccountTriggerHandler.cls — Step by Step

This class does two jobs in one file (merged for simplicity from the original 2-class version): it holds the business rule, and it decides when the trigger should apply it.

### 2.1 — Constants block

```
public static final String TIER_ENTERPRISE = 'Enterprise';
public static final String TIER_LARGE      = 'Large';
public static final String TIER_MEDIUM    = 'Medium';
public static final String TIER_SMALL     = 'Small';
```

```
public static final String RATING_HOT  = 'Hot';
public static final String RATING_WARM = 'Warm';
public static final String RATING_COLD = 'Cold';
```

**What:** every picklist value the class ever writes is declared once, as a constant, instead of typed out as a raw string everywhere. **Why:** if you typo 'Enterprise' in one method but not another, Apex won't catch it at compile time (they're just strings) — a constant makes that typo impossible, and if management ever renames a tier, you change it in exactly one line.

### 2.2 — calculateTier(): the core business rule

```
public static String calculateTier(Decimal annualRevenue, Integer numberOfEmployees) {
    Decimal revenue = annualRevenue == null ? 0 : annualRevenue;
    Integer employees = numberOfEmployees == null ? 0 : numberOfEmployees;

    if (revenue >= 1000000000 || employees >= 10000) return TIER_ENTERPRISE;
    if (revenue >= 100000000 || employees >= 1000)   return TIER_LARGE;
    if (revenue >= 1000000 || employees >= 100)      return TIER_MEDIUM;
    return TIER_SMALL;
}
```

- **Null-safety first:** the ternary lines convert a null AnnualRevenue/NumberOfEmployees into 0 before any comparison. In your real CSV data this matters — any row where those columns were blank on import would otherwise cause null >= 1000000000, which is valid Apex but always evaluates false in a way that's easy to reason about wrong; treating it as 0 makes the intent explicit.

- **Order is deliberate:** Enterprise is checked first, then Large, then Medium. Because each rule is an OR of two conditions, and the tiers are meant to be mutually exclusive, checking from the highest tier down and returning immediately on the first match is both correct and the cheapest possible implementation — no need to evaluate the lower rules once a higher one matches.
- **Pure function:** no SOQL, no DML, no reference to `Trigger.new`. It only depends on its two parameters. That's what makes §4's boundary tests possible without inserting a single record.

### 2.3 — *calculateRating(): the Tier → Rating map*

```
public static String calculateRating(String tier) {
    if (tier == TIER_ENTERPRISE || tier == TIER_LARGE) return RATING_HOT;
    if (tier == TIER_MEDIUM) return RATING_WARM;
    return RATING_COLD;
}
```

Straight lookup table implemented as if/else rather than a Map — for four fixed values a Map buys nothing and if/else reads more obviously top-to-bottom. Falling through to `RATING_COLD` for anything unrecognised (including null) means this method can never throw, which matters because it's called from inside a trigger — an uncaught exception there would block the entire save.

### 2.4 — *applyTierAndRating(): the shared, bulk-safe entry point*

```
public static Integer applyTierAndRating(List<Account> accounts) {
    Integer changedCount = 0;
    for (Account acc : accounts) {
        String newTier = calculateTier(acc.AnnualRevenue, acc.NumberOfEmployees);
        String newRating = calculateRating(newTier);
        Boolean changed = acc.Account_Tier__c != newTier || acc.Rating != newRating;
        if (changed) {
            acc.Account_Tier__c = newTier;
            acc.Rating = newRating;
            changedCount++;
        }
    }
    return changedCount;
}
```

- **Takes a List, not a single Account.** This is the single habit that separates code that survives a Data Loader import of 100,000 rows from code that only ever worked in the UI with one record at a time. Both the trigger (§2.5/2.6) and the batch job (§3) call this exact same method.
- **The changed check is what avoids unnecessary updates.** It computes what the values **SHOULD** be, compares against what they **ALREADY** are, and only writes (and counts) a change if something is actually different. This is what your batch test later measures numerically (80 of 160 updated, not 160).
- **Returns a count, not void.** The batch job doesn't use this return value directly (it does its own counting in §3, since it also needs `Database.update`'s per-record success/failure), but exposing it here makes the method independently useful and testable — see the service-level test in §4.3.

### 2.5 — *handleBeforeInsert(): the simple case*

```
public static void handleBeforeInsert(List<Account> newAccounts) {
    applyTierAndRating(newAccounts);
}
```

Every new Account gets a Tier/Rating unconditionally — there's no "old version" to compare against on an insert, so there's no decision to make about whether to recalculate. This is what fired for every one of your 100,000 imported rows if the trigger was active during the CSV load.

### 2.6 — *handleBeforeUpdate(): the selective case*

```
public static void handleBeforeUpdate(List<Account> newAccounts, Map<Id, Account> oldMap) {
    List<Account> toRecalculate = new List<Account>();
    for (Account acc : newAccounts) {
        Account oldAcc = oldMap.get(acc.Id);
```

```

        if (acc.AnnualRevenue != oldAcc.AnnualRevenue || acc.NumberOfEmployees != oldAcc.NumberOfEmployees) {
            toRecalculate.add(acc);
        }
    }
    if (!toRecalculate.isEmpty()) {
        applyTierAndRating(toRecalculate);
    }
}

```

- **Trigger.oldMap** is only available in update/delete contexts (there's no "before" version of a brand-new record), which is why this method takes it as a parameter and `handleBeforeInsert` doesn't.
- **The if-check is the whole point of this method existing separately from `handleBeforeInsert`:** editing an Account's Phone number, Website, or any other field should NOT re-run tier math. Only a change to the two fields that actually drive the rule (AnnualRevenue, NumberOfEmployees) triggers a recalculation.
- **Building a filtered list first, then calling `applyTierAndRating` once** (rather than calling it inside the loop per record) keeps the whole operation bulk-safe — one pass over the incoming records, one call into the shared logic.

### 3. AccountTrigger.trigger — Step by Step

```
trigger AccountTrigger on Account (before insert, before update) {
    if (Trigger.isBefore && Trigger.isInsert) {
        AccountTriggerHandler.handleBeforeInsert(Trigger.new);
    }
    if (Trigger.isBefore && Trigger.isUpdate) {
        AccountTriggerHandler.handleBeforeUpdate(Trigger.new, Trigger.oldMap);
    }
}
```

- **"before insert, before update"** in the trigger declaration is what makes the two if blocks below ever reachable — a trigger only receives the events it's declared for. Deliberately not including after events, because nothing in this task needs them.
- **Before, not after:** in a before-trigger, changes made to fields on records in `Trigger.new` are automatically folded into the save that's already happening. No second DML statement, no re-firing the trigger. If this were an after-trigger, setting `Account_Tier__c` would require a separate update call inside the trigger — an extra DML statement per save, and a real risk of infinite recursion if not carefully guarded.
- **Trigger.new vs Trigger.oldMap:** `Trigger.new` is a `List` (available in every context); `Trigger.oldMap` is a `Map` keyed by `Id` (only available on update/delete, which is exactly why it's only passed into `handleBeforeUpdate`).
- **Zero business logic in this file.** If you ever need a second automation on `Account` (a Flow, another trigger requirement), you'll never accidentally end up with two competing `Account` triggers racing in an undefined order — there's exactly one, and it just dispatches.

### 4. AccountTierBatch.cls — Step by Step

This is the piece that matters most for your actual 100,000-row dataset: it's what backfills `Account_Tier__c/Rating` on records that already existed before (or independently of) any trigger activity.

#### 4.1 — Class declaration and state

```
public with sharing class AccountTierBatch implements Database.Batchable<SObject>, Database.Stateful {

    public Map<String, Integer> tierCounts = new Map<String, Integer>{
        AccountTriggerHandler.TIER_ENTERPRISE => 0,
        AccountTriggerHandler.TIER_LARGE      => 0,
        AccountTriggerHandler.TIER_MEDIUM    => 0,
        AccountTriggerHandler.TIER_SMALL     => 0
    };
    public Integer recordsProcessed = 0;
    public Integer recordsUpdated = 0;
```

- **Database.Batchable<SObject>** is the interface that forces this class to implement exactly three methods: `start()`, `execute()`, `finish()` — the three phases every batch job goes through.
- **Database.Stateful** is easy to miss and easy to get wrong: WITHOUT it, `tierCounts/recordsProcessed/recordsUpdated` would silently reset to their initial values before every single `execute()` call, and `finish()` would only ever report numbers from the very last chunk out of ~50. WITH it, these three variables persist across the whole job, so `finish()` sees the true grand total.
- **tierCounts starts pre-populated** with all four keys at 0, rather than being built up dynamically, so that a tier with zero matching `Accounts` still shows up as "0" in the summary instead of being missing from the map entirely.

#### 4.2 — start(): defining the scope of the job

```
public Database.QueryLocator start(Database.BatchableContext bc) {
    return Database.getQueryLocator([
        SELECT Id, AnnualRevenue, NumberOfEmployees, Account_Tier__c, Rating
```

```

        FROM Account
    });
}

```

Runs exactly once, at the very start of the job. Returning a **Database.QueryLocator** (instead of, say, a `List<Account>`) is the specific design choice that lets this scale to your 100,000 rows safely — a `QueryLocator` can stream up to 50 million records without ever holding them all in memory at once. The query only selects the five fields actually needed (not `SELECT *`, which doesn't even exist in SOQL, but the general principle of selecting only what you use).

### 4.3 — *execute()*: the work, done in safe chunks

```

public void execute(Database.BatchableContext bc, List<Account> scope) {
    List<Account> toUpdate = new List<Account>();
    for (Account acc : scope) {
        String tier = AccountTriggerHandler.calculateTier(acc.AnualRevenue, acc.NumberOfEmployees);
        String rating = AccountTriggerHandler.calculateRating(tier);
        recordsProcessed++;
        tierCounts.put(tier, tierCounts.get(tier) + 1);

        if (acc.Account_Tier__c != tier || acc.Rating != rating) {
            acc.Account_Tier__c = tier;
            acc.Rating = rating;
            toUpdate.add(acc);
        }
    }
    if (!toUpdate.isEmpty()) {
        List<Database.SaveResult> results = Database.update(toUpdate, false);
        for (Database.SaveResult sr : results) {
            if (sr.isSuccess()) recordsUpdated++;
        }
    }
}

```

- **Called once per chunk ("scope")**, not once per record and not once for the whole job. If you call `Database.executeBatch(new AccountTierBatch(), 2000)`, this method runs roughly 50 times for your 100,000 records, each time with up to 2,000 Accounts in scope.
- **The counting happens for every record** (`recordsProcessed++` and `tierCounts.put(...)` run unconditionally), but the record only gets added to `toUpdate` — and therefore only costs a DML row — if it's actually wrong. This is the exact same "compare before writing" idea as §2.4, just re-implemented here because the batch also needs to count successes from `Database.SaveResult`, which `applyTierAndRating()` doesn't return.
- **`Database.update(toUpdate, false)`** — the `false` is `allOrNone`. With it `false`, if one record in a 2,000-record chunk fails (e.g. a validation rule conflict), the other 1,999 still save; only that one record's failure is skipped. With the default `true`, a single bad record would roll back the entire chunk.
- **`sr.isSuccess()`** is checked per record so `recordsUpdated` reflects records that ACTUALLY saved, not just records that were attempted.

### 4.4 — *finish()*: the summary

```

public void finish(Database.BatchableContext bc) {
    System.debug('Processed: ' + recordsProcessed + ', Updated: ' + recordsUpdated);
    System.debug('Tier counts: ' + tierCounts);
}

```

Runs exactly once, after every chunk has completed. Because of `Database.Stateful` (§4.1), the numbers printed here are the true totals across all ~50 chunks, not just the last one. In a production org you'd typically also send an email here (`Messaging.SingleEmailMessage`) — for this task, Debug Logs plus the verification query in §6 are sufficient evidence.

## 5. AccountTriggerHandlerTest.cls — Step by Step

### 5.1 — testCalculateTierBoundaries()

```
System.assertEquals('Small', AccountTriggerHandler.calculateTier(null, null));
System.assertEquals('Medium', AccountTriggerHandler.calculateTier(1000000, 0));
System.assertEquals('Large', AccountTriggerHandler.calculateTier(0, 1000));
System.assertEquals('Enterprise', AccountTriggerHandler.calculateTier(1, 10000));
```

No DML at all in this test — it calls the pure calculateTier() method directly with values chosen to sit EXACTLY on each threshold. calculateTier(1, 10000) in particular proves the OR logic: revenue of just \$1 combined with 10,000 employees still correctly returns Enterprise.

### 5.2 — testCalculateRatingMapping()

Same idea, applied to calculateRating() — four assertions, one per tier, confirming the Hot/Hot/Warm/Cold mapping from the requirements table exactly.

### 5.3 — testBulkInsertSetsTierAndRating()

```
List<Account> accounts = new List<Account>();
for (Integer i = 0; i < 200; i++) {
    accounts.add(new Account(Name = 'Acc ' + i, AnnualRevenue = i * 6000000, NumberOfEmployees = i * 60));
}
Test.startTest();
insert accounts;
Test.stopTest();

for (Account acc : [SELECT AnnualRevenue, NumberOfEmployees, Account_Tier__c, Rating FROM Account]) {
    String expectedTier = AccountTriggerHandler.calculateTier(acc.AnnualRevenue, acc.NumberOfEmployees);
    System.assertEquals(expectedTier, acc.Account_Tier__c);
    System.assertEquals(AccountTriggerHandler.calculateRating(expectedTier), acc.Rating);
}
```

- **200 records in ONE insert statement** (not 200 separate inserts) is what actually tests bulkification — it proves the trigger fires once for the whole list and correctly processes every record in that single invocation, the same way your CSV import sent Accounts to the trigger in batches, not one row at a time.
- **The loop that builds the list** ( $i * 6000000$ ,  $i * 60$ ) deliberately sweeps values across every tier boundary in one dataset, rather than testing only one tier.
- **Test.startTest()/Test.stopTest()** wrap only the DML — this resets governor limits right before the insert, isolating the insert's cost from whatever limits the list-building loop above it used.
- **The assertion loop re-derives the expected value** using the same calculateTier/calculateRating methods being tested, rather than hardcoding 200 expected values by hand — this is what makes the test both correct and easy to extend if you add more records later.

### 5.4 — testUpdateOnlyRecalculatesWhenDriverFieldsChange()

```
Account acc = new Account(Name = 'Test', AnnualRevenue = 0, NumberOfEmployees = 0);
insert acc; // Small/Cold

acc.Phone = '555-1234'; // unrelated field
update acc;
Account unchanged = [SELECT Account_Tier__c FROM Account WHERE Id = :acc.Id];
System.assertEquals('Small', unchanged.Account_Tier__c);

acc.AnnualRevenue = 2000000000; // should jump to Large/Hot
update acc;
Account changedAcc = [SELECT Account_Tier__c, Rating FROM Account WHERE Id = :acc.Id];
System.assertEquals('Large', changedAcc.Account_Tier__c);
System.assertEquals('Hot', changedAcc.Rating);
```

Two updates against the SAME record, back to back. The first (Phone) proves the negative case: unrelated field edits do NOT touch Tier/Rating — this is what "unnecessary updates avoided" means for the real-time path. The second (AnnualRevenue) proves the positive case AND that a Tier change correctly cascades into a Rating

change in the same save.

## 6. AccountTierBatchTest.cls — Step by Step

### 6.1 — @TestSetup: building a realistic "mixed" dataset

```
@TestSetup
static void setup() {
    List<Account> accounts = new List<Account>();
    for (Integer i = 0; i < 40; i++) accounts.add(new Account(Name='Ent '+i, AnnualRevenue=2000000000, NumberOfEmployees=2000));
    for (Integer i = 0; i < 40; i++) accounts.add(new Account(Name='Lg '+i, AnnualRevenue=200000000, NumberOfEmployees=200));
    for (Integer i = 0; i < 40; i++) accounts.add(new Account(Name='Md '+i, AnnualRevenue=5000000, NumberOfEmployees=500));
    for (Integer i = 0; i < 40; i++) accounts.add(new Account(Name='Sm '+i, AnnualRevenue=0, NumberOfEmployees=0));
    insert accounts; // trigger sets these all correctly on insert

    List<Account> toCorrupt = new List<Account>();
    for (Integer i = 0; i < accounts.size(); i++) {
        if (Math.mod(i, 2) == 0) {
            accounts[i].Account_Tier__c = 'Small';
            accounts[i].Rating = 'Cold';
            toCorrupt.add(accounts[i]);
        }
    }
    update toCorrupt;
}
```

- **@TestSetup runs once** and Salesforce automatically rolls the org back to this exact snapshot before EACH @isTest method in the class runs — so both test methods below see the same starting 160 records without interfering with each other.
- **The first insert is intentionally "too correct":** because AccountTrigger is active, all 160 records get the right Tier/Rating the moment they're inserted, which on its own would leave nothing for the batch to fix.
- **The second block deliberately corrupts every other record** — but only Account\_Tier\_\_c/Rating, never AnnualRevenue/NumberOfEmployees. Look back at §2.6: handleBeforeUpdate() only recalculates when the driver fields change. Because this update doesn't touch them, the trigger lets the corrupted values through untouched — which is exactly what lets these tests prove the BATCH (not the trigger) is what fixes them.
- **Math.mod(i, 2) == 0** corrupts exactly 80 of the 160 records (one in every tier group), giving a precise, predictable number for §6.3's assertion.

### 6.2 — testBatchFixesAllTiers()

```
Test.startTest();
Database.executeBatch(new AccountTierBatch(), 50);
Test.stopTest();

Map<String, Integer> counts = new Map<String, Integer>();
for (AggregateResult ar : [SELECT Account_Tier__c tier, COUNT(Id) total FROM Account GROUP BY Account_Tier__c]) {
    counts.put((String) ar.get('tier'), (Integer) ar.get('total'));
}
System.assertEquals(40, counts.get('Enterprise'));
System.assertEquals(40, counts.get('Large'));
System.assertEquals(40, counts.get('Medium'));
System.assertEquals(40, counts.get('Small'));
```

- **Test.startTest()/stopTest() around Database.executeBatch** is what makes the batch run SYNCHRONOUSLY for the test — normally a batch job runs asynchronously and you'd check Setup → Apex Jobs, but inside a test, stopTest() doesn't return until start/execute/finish have all completed, so the assertions right after it can immediately query final results.
- **Scope size 50** here (vs. 2,000 in production) just means the test batch runs in more, smaller chunks — useful for exercising the chunking logic itself even with a small 160-record test dataset.



- **An AggregateResult / GROUP BY query** is the standard SOQL pattern for a "count per category" summary — exactly what Requirement 3's expected output table needs.

### 6.3 — *testBatchSkipsAlreadyCorrectRecords()*

```
AccountTierBatch job = new AccountTierBatch();  
Test.startTest();  
Database.executeBatch(job, 50);  
Test.stopTest();
```

```
System.assertEquals(160, job.recordsProcessed);
```

```
System.assertEquals(80, job.recordsUpdated); // only the corrupted half needed fixing
```

This is the test that actually **PROVES** the efficiency claim, rather than just trusting the code comment. By keeping a reference to the batch instance (job) before calling executeBatch, and by using Database.Stateful (§4.1) so its counters survive the whole run, the test can assert the exact number updated (80) against the exact number processed (160) — demonstrating that the other 80 records, which were already correct, never triggered a DML write at all.

## 7. Running This Against Your Imported 100,000-Row Dataset

- 1 **Confirm current state.** Open Developer Console → Query Editor (or Execute Anonymous) and run: `SELECT COUNT(Id), Account_Tier__c FROM Account GROUP BY Account_Tier__c` — if every row already has a non-blank tier, the trigger was active during your import and already did the work; if they're all blank, the batch has real work to do.
- 2 **Deploy all 5 files** in this order if you haven't already: `AccountTriggerHandler.cls` → `AccountTrigger.trigger` → `AccountTierBatch.cls` → both test classes.
- 3 **Run the batch.** Debug → Open Execute Anonymous Window: `Database.executeBatch(new AccountTierBatch(), 2000);` — for 100,000 records at scope size 2,000, expect ~50 chunks. Watch Setup → Apex Jobs for completion.
- 4 **Read the summary from Debug Logs** (`finish()` writes `recordsProcessed/recordsUpdated/tierCounts` there), or re-run the GROUP BY query from step 1 for the same numbers straight from the data.
- 5 **Run the tests.** Setup → Apex Test Execution → Select All (or right-click each class in Developer Console → Run Test) → note the coverage percentage for your submission.

## 8. Answers to the Required Written Explanation (Deliverable 8)

### *How the existing 100,000 Account records were processed*

Via **AccountTierBatch**, a `Database.Batchable` class. `start()` streams every `Account` through a `Database.QueryLocator`; `execute()` runs against 2,000-record chunks, keeping the whole run within governor limits regardless of dataset size.

### *How Account Tier is calculated*

**AccountTriggerHandler.calculateTier()** checks `AnnualRevenue` and `NumberOfEmployees` against the Enterprise → Large → Medium thresholds from highest to lowest, returning on the first match, defaulting to Small. Nulls are treated as zero.

### *How Rating is kept synchronized with Account Tier*

Both the real-time path (`AccountTrigger` → `AccountTriggerHandler`) and the bulk path (`AccountTierBatch`) call the exact same **calculateRating(tier)** method — there is only one place the mapping is defined, so it cannot drift between the two paths.

### *How the implementation handles multiple records efficiently*

Every method operates on Lists, never single records; the batch uses a `QueryLocator` and chunked `execute()` calls instead of one giant query/update; `Database.update(list, false)` uses `allOrNone=false` so one bad record never rolls back an entire chunk.

### *How unnecessary record updates are avoided*

Both `applyTierAndRating()` (real-time path) and `execute()` (batch path) compute the correct Tier/Rating and compare it against the record's current value BEFORE deciding to write anything.

`testBatchSkipsAlreadyCorrectRecords` proves this numerically: of 160 test records, exactly 80 (the deliberately

corrupted half) triggered an update — the other 80 were left untouched.

— End of report —